BIO504T SUBJECTIVES

MID TERM- 100%

BIOCHEMISTRY I

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SUBJECTIVES

What is titration curve of glycine? (short)

The titration curve of glycine illustrates its amino acid behavior as it undergoes titration, showing distinct pH changes at its amino and carboxyl groups. The curve displays a characteristic dip around its pKa values, indicating protonation or deprotonation.

What is titration curve of glycine? (long)

The titration curve of glycine reveals key information: pKa values of its ionizing groups (2.34 for -COOH, 9.60 for -NH3+), indicating acidic and basic strengths. Two buffering regions emerge: one near pH 2.6 and another around pH 9.60. Glycine is an effective buffer within these zones. Notably, at physiological pH (around 7.4), glycine lacks buffering capacity. The curve also illustrates glycine's net charge at specific pH values, highlighting its dipolar form at pH 5.97, fully ionized but electrically neutral.

What is starch and its structure?

Starch is a complex carbohydrate composed of glucose units linked by alpha-1,4-glycosidic bonds in linear chains (amylose) and branched structures with alpha-1,6-glycosidic bonds (amylopectin). This polysaccharide is a major storage form of energy in plants, found in grains, tubers, and seeds. The unique branching of amylopectin allows for efficient energy storage and subsequent enzymatic hydrolysis for glucose release.

Define Renaturation of ribonuclease A:

Renaturation of ribonuclease A involves restoring its native structure and biological activity after denaturation, often induced by heat, extreme pH, or denaturing agents. When returned to conditions favoring native conformation stability, the protein refolds, exemplified by the denaturation-renaturation process in concentrated urea solutions.

Cooperative binding of oxygen:

Cooperative binding of oxygen means that the binding of oxygen to one heme group enables an oxygen binding to the second heme group of the same hemoglobin molecule. Conversely, the oxygen release from one heme group facilitates the release of oxygen from other heme groups in the same hemoglobin molecule.

Give Maltose Sucrose and lactose hydrolysis product:

1. Maltose - 2 molecules of glucose on hydrolysis
2. Lactose – 1 glucose & 1 galactose on hydrolysis
3. Sucrose – 1 glucose & 1 fructose on hydrolysis

Lactose intolerance and its symptoms

Lactose intolerance is the inability to digest lactose, a sugar found in milk and dairy products. It results from low lactase levels or intestinal injury. Symptoms include abdominal distention, cramping, pain, and watery diarrhea due to osmotic effects and fluid influx in the small intestine.

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Define Induced fit:

The induced fit model in biochemistry describes how enzymes and substrates interact during a chemical reaction. It emphasizes that the enzyme's active site isn't a rigid structure but can change its shape slightly when a substrate binds. This dynamic adjustment optimizes the fit between the enzyme and substrate, enhancing the catalytic efficiency of the enzyme in facilitating the chemical reaction. It's like a molecular handshake that fine-tunes for better coordination, ensuring the most effective conversion of substrates into products.

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Structural proteins and three structural proteins name:

Structural proteins provide mechanical support and shape to cells and tissues. Examples: Collagen – Elastin – Keratin

Define Four levels of proteins:

Proteins exhibit four levels of structural organization: Primary structure involves the linear sequence of amino acids in a polypeptide chain. Secondary structure results from interactions such as alpha-helices and beta-sheets, folding due to hydrogen bonding. Tertiary structure represents the overall three-dimensional arrangement of a single polypeptide chain. Quaternary structure involves the interaction of multiple polypeptide chains in a protein complex. These hierarchical levels collectively determine a protein's function and are crucial for its stability and biological activity.

Factor of influence free energy of protein?

The free energy of a protein molecule is influenced by: 1= The hydrophobicity, 2= Hydrogen bonds, 3= Electrostatic interactions, 4= The conformational entropy due to the restricted motion

Or

1. **Hydrophobicity:** Hydrophobic interactions contribute to protein folding by minimizing exposure of hydrophobic residues to aqueous environments.
2. **Hydrogen Bonds:** Hydrogen bonds stabilize protein structure through interactions between polar groups in the peptide backbone or side chains.
3. **Electrostatic Interactions:** Electrostatic forces between charged amino acids influence protein folding and stability.

4. **Conformational Entropy:** Restricted motion affects conformational entropy, influencing the free energy landscape of protein folding.

What is D and L isomerism with glucose example?

D.L isomerism refers to the spatial arrangement of isomers based on the orientation of asymmetric carbons. In glucose, the D isomer has the hydroxyl group on the right at the highest-priority chiral carbon, while the L isomer has it on the left.

Primary structure of protein:

The primary structure of a protein refers to its linear sequence of amino acids, determined by the specific order of peptide bonds. It serves as the foundation for higher-order structural configurations in proteins.

What are the Example of amino sugar:

1= D -glucosamine, a constituent of hyaluronic acid, 2= D -galactosamine (also known as chondrosamine), a constituent of chondroitin, 3= D –mannosamine

Write a note on the composition of protein:

Proteins, essential biological macromolecules, consist of amino acid chains linked by peptide bonds. Comprising carbon, hydrogen, oxygen, nitrogen, and sometimes sulfur, proteins exhibit diverse structures and functions crucial to life. The 20 different amino acids, each with a unique side chain, contribute to the varied composition of proteins. This structural diversity enables proteins to fulfill roles in enzymatic catalysis, cell structure, signaling, immune response, and numerous other biological processes.

What is role proton bind with heme group in tissues and lung?

Protons binding to the heme group in tissues facilitate oxygen release from hemoglobin, promoting oxygen delivery to cells. In the lungs, protons binding to heme assist in oxygen uptake, optimizing the respiratory cycle and ensuring efficient gas exchange.

What is disaccharide give two examples:

A disaccharide (also called a double sugar or blase) is the sugar formed when two monosaccharides (simple sugars) are joined by glycosidic linkage. Like monosaccharides, disaccharides are soluble in water. Three common examples are sucrose, lactose, and maltose.

Function of auxiliary protein:

Auxiliary proteins play supporting roles, aiding the function of other proteins. They may assist in folding, transport, or stabilization. Examples include chaperones aiding proper protein folding and cofactors assisting enzymes in catalyzing reactions.

The domain function of proteins:

Protein domains are distinct, modular units with specific functions within a larger protein. They contribute to the protein's overall functionality by mediating interactions, binding ligands, or participating in enzymatic activities, enhancing the protein's adaptability and versatility in diverse cellular processes.

Importance of glucose:

Glucose is a vital energy source for the human body, serving as the primary fuel for cellular activities. It plays a crucial role in metabolism, providing energy for essential processes such as respiration and ATP production. Additionally, glucose is integral to maintaining blood sugar levels, supporting brain function, and contributing to various biosynthetic pathways essential for growth, repair, and overall physiological balance.

Explain Secondary structure of protein:

The secondary structure of a protein refers to local folding patterns, predominantly alpha-helices and beta-sheets. These structures arise from hydrogen bonding between amino acids within the polypeptide chain. Alpha-helices form a coiled structure, while beta-sheets involve extended strands. Secondary structure contributes to the overall three-dimensional arrangement of a protein, influencing its stability and functionality. These repetitive folding patterns are critical in determining a protein's shape and biological activity.

What is Hsp60:

Hsp60, or Heat Shock Protein 60, is a molecular chaperone found in cells. It assists in protein folding, preventing misfolding or aggregation, especially during cellular stress. Hsp60 is involved in maintaining cellular homeostasis and supporting proper protein function.

Differentiate between Homopollysaccharides and heteropollysaccharides:

Homopolysaccharides are polysaccharides composed of repeating units of the same type of monosaccharide, such as cellulose (composed of glucose units). Heteropolysaccharides consist of different types of monosaccharides in their repeating units, such as glycosaminoglycans (e.g., heparin) or peptidoglycans, exhibiting structural diversity and varied functions in biological systems.

Equation for pressure:

Pressure (P) is defined as force (F) per unit area (A). The equation for pressure is given by:

$P=F/A$

where: P = Pressure, F = Force, A = Area.

What are Chaperons and write its main function?

Chaperones are proteins that assist in the folding, assembly, and transport of other proteins. Their main function is to prevent misfolding or aggregation, ensuring the correct three-dimensional structure of proteins, promoting cellular integrity, and preventing protein-associated diseases.

Non-essential amino acids:

Non-essential amino acids are those that the human body can synthesize internally and are not required in the diet. While still crucial for protein synthesis. For example: glutamic acid, glutamine, glycine, proline, serine, and tyrosine.

Beta confirmation of proteins:

The beta conformation in proteins involves the polypeptide chain adopting a pleated sheet structure, where adjacent strands run in opposite directions and are stabilized by hydrogen bonds between their carbonyl and amino groups. Beta sheets contribute to protein stability and structure, forming an essential component of the secondary structure. They play crucial roles in protein folding and function in various biological processes.

How mis-folding of proteins occur?

Misfolding of proteins can occur due to genetic mutations, environmental factors, or errors during synthesis. Inappropriate folding leads to non-functional or abnormal structures, disrupting cellular processes. Misfolded proteins may aggregate, contributing to various neurodegenerative diseases and proteinopathies.

How gluconic, glucuronic acids form glucose:

Gluconic and glucuronic acids from glucose through oxidation reactions. Gluconic acid results from the oxidation of glucose's aldehyde group, while glucuronic acid forms by oxidizing one of the primary alcohol groups in glucose, typically via enzymatic processes.

Properties of proteins:

Proteins exhibit diverse properties crucial for biological functions, including enzymatic catalysis, structural support, and cellular signaling. Their properties depend on the amino acid sequence, folding patterns, and interactions. Proteins are sensitive to environmental factors, such as pH and temperature, and their functionality relies on proper folding and structural integrity.

Calculate pI for glutamate:

pI=1/2(pKa1+pKa2)

For glutamate, the pKa values are approximately as follows:

pKa1≈2.2 (for the carboxyl group)

pKa2≈4.2 (for the second carboxyl group)

pKa3≈9.7 (for the amino group)

Now, to calculate the pI for glutamate:

pI=1/2(pKa1+pKa2)

pI=1/2(2.2+4.2)

pI=1/2×6.4

pI=3.22

What are epimers:

Two sugars which differ from one another only in configuration around a single carbon atom are termed epimers. Glucose and Galactose are an example of an epimeric pair which differ only with respect to C4.

Define Osazone:

crystalline derivatives of sugars called osazones • They are obtained by adding a mixture of phenyl hydrazine hydrochloride and sodium acetate to a sugar solution and heating in water bath for 45 mins.

pI of glycine determination:

The pI is simply the arithmetic mean of the two pKa values.

pI = ½ (pK1+pK2)

= ½ (2.34+9.60) = 5.97

Write a note on Allosteric effect:

Allosteric effects in hemoglobin regulation involve factors like pH, pCO_2, and 2,3-bisphospho-phoglycerate, collectively termed allosteric effectors. These molecules interact at sites on the hemoglobin molecule distinct from the oxygen-binding sites, influencing its reversible oxygen binding. Myoglobin, unlike hemoglobin, is not influenced by these effectors. Allosteric effects allow hemoglobin to adapt its oxygen-binding affinity to the physiological needs of tissues, enabling efficient gas transport and exchange.

Action of acids on carbohydrates:

Acids catalyze the hydrolysis of carbohydrates through protonation of functional groups, initiating chemical reactions that break glycosidic bonds. In this process, acids donate protons, destabilizing the bonds holding together sugar molecules. This acid-catalyzed hydrolysis results in the conversion of complex carbohydrates into simpler sugars, facilitating their absorption and utilization. The reaction plays a crucial role in digestive processes and chemical transformations in various biological and industrial contexts.

What is the function of sorbitol?

Sorbitol is a sugar alcohol found in fruits and plants with diuretic, laxative and cathartic property. Unabsorbed sorbitol retains water in the large intestine through osmotic pressure thereby stimulating peristalsis of the intestine and exerting its diuretic, laxative and cathartic effect. Sorbitol is also used as sweetener.

Dissociation of amino group of Alanine":

In aqueous solution, the amino group of alanine undergoes dissociation, releasing a proton (H+) and forming its conjugate base. This process involves the ionization of the amino group, making alanine amphiprotic and contributing to its role in acid-base chemistry.

What are the major classes or carbohydrates?

Carbohydrates are broadly categorized into monosaccharides (single sugar units), disaccharides (two linked monosaccharides), and polysaccharides (long chains of monosaccharides). Monosaccharides include glucose, fructose, and galactose. Examples of disaccharides are sucrose, lactose, and maltose. Polysaccharides include starch, glycogen, and cellulose.

Free energy equation:

The Gibbs free energy equation is $\Delta G = \Delta H - T\Delta S$, where ΔG is the change in free energy, ΔH is the change in enthalpy, ΔS is the change in entropy, and T is the absolute temperature in Kelvin. It describes the spontaneity of a reaction under given conditions.

Characteristics of glycine:

Glycine is the simplest amino acid, with a hydrogen side chain. It is nonpolar and possesses a unique role as an inhibitory neurotransmitter in the central nervous system. As a building block of proteins, glycine plays a crucial role in collagen formation. Its flexibility allows it to fit into tight protein structures. Additionally, glycine has an essential role in bile salt conjugation and serves as a neurotransmitter precursor.

Categorize isomers on the bases of classification:

Isomers can be categorized into structural isomers, which differ in the arrangement of atoms; stereoisomers, with identical connectivity but different spatial arrangement; and conformational isomers, variations in rotation around single bonds. Stereoisomers further include geometric and optical isomers based on spatial orientation.

Properties of side chain of glutamate and asparagine:

The side chain of glutamate is a negatively charged carboxyl group, contributing to its acidic nature. Asparagine's side chain contains an amide group, imparting a polar and neutral character. Both side chains play distinct roles in protein structure and function.

Geometric isomers in details:

Geometric isomers, exemplified by maleic and fumaric acid, exhibit distinct arrangements of substituents around a nonrotating double bond. Unlike conformational isomers, these well-defined compounds cannot interconvert without breaking covalent bonds. The cis-trans configuration of geometric isomers imparts unique chemical properties, allowing separation.

This structural dissimilarity arises from spatial orientation, influencing reactivity and properties, making geometric isomerism crucial in understanding and manipulating molecular behavior.

Describe transport protein and contractile protein:

1= Transport proteins facilitate the movement of substances across cell membranes or within cells, ensuring proper cellular function. Examples include ion channels, carriers, and pumps.

2= Contractile proteins, like actin and myosin, are essential for muscle contraction. They interact to generate mechanical force and movement, vital for muscle function.

Both protein types play crucial roles in cellular and physiological processes, highlighting their significance in maintaining cellular homeostasis and supporting various biological functions.

Negatively Charged (Acidic) R Groups

These amino acids are proton donors • At neutral pH, the side chains of these amino acids are fully ionized, containing a negatively charged carboxylate group • (--COO-) • Therefore these amino acids are negatively charged at phyisologic pH • The two amino acids having R groups with a net negative charge at pH 7.0 are aspartate and glutamate, each of which has a second carboxyl group.

Protein involve in respiration:

Proteins participate in respiration as enzymes within metabolic pathways. Enzymes, like those in the electron transport chain, facilitate the transfer of electrons, leading to the production of ATP during oxidative phosphorylation, a central process in cellular respiration.

Free energy change at constant pressure and equation:

Free energy change (ΔG) is the energy available to do work at constant pressure, indicating a reaction's spontaneity., Equation $\Delta G = \Delta H - T\Delta S$.

What are ketogenic amino acids?

Ketogenic amino acids are those that can be converted into ketone bodies through catabolic pathways. Leucine and lysine are primarily ketogenic, contributing to the production of ketones, which can be utilized as an energy source, especially during fasting or low-carbohydrate conditions.

What is meant by protein folding?

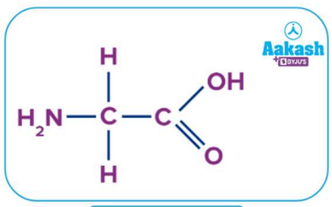
Protein folding refers to the process by which a linear chain of amino acids assumes its three-dimensional, functional structure. It involves intricate spatial arrangements, driven by interactions such as hydrogen bonding, hydrophobic forces, and disulfide bonds, ensuring proper protein functionality.

Explain PKa?

pKa is the negative logarithm (base 10) of the acid dissociation constant (Ka), representing the strength of an acid in solution. It quantifies the tendency of an acid to donate protons. Lower pKa values indicate stronger acids. In biochemistry, it is crucial for understanding the ionization state of functional groups, especially in amino acids, influencing protein structure and function.

Structure of glycine:

Glycine is the simplest amino acid with a hydrogen atom as its side chain. Its molecular formula is C2H5NO2, and its structure consists of a central carbon atom bonded to a hydrogen atom, an amino group (NH2), and a carboxyl group (COOH).



Structure of Glycine

Define saccharides and give two examples:

Saccharides are organic compounds, commonly known as sugars, that serve as essential carbohydrates in living organisms. Examples include glucose, a primary energy source, and fructose, found in fruits. Saccharides play crucial roles in cellular energy production and structural components.

Explain Amyloidosis:

Amyloidosis is a rare disorder where abnormal proteins, called amyloids, accumulate in organs and tissues, disrupting their normal function. These misfolded proteins can deposit in various areas, leading to organ damage. Symptoms vary depending on affected organs. Diagnosis involves tissue biopsy. Treatment targets underlying conditions or symptoms, including chemotherapy, stem cell transplants, or supportive measures.

Isomers and its type?

Isomers are compounds with the same molecular formula but different arrangements of atoms, yielding distinct chemical structures. Types include structural isomers (different connectivity), geometric isomers (differ in spatial arrangement), and stereoisomers (differ in spatial orientation, including enantiomers and diastereomers).

Lactose and its properties:

Lactose is a disaccharide sugar found in milk. Composed of glucose and galactose, it requires the enzyme lactase for digestion. Lactose intolerance arises when lactase is deficient. It serves as a source of energy and contributes to the sweetness of milk.

Abbreviation of Methionine, Arginie, Aanine, asparagine, Glycine?

1= Methionine: Met=M

2= Arginine: Arg=R

3= Alanine: Ala=A

4= Asparagine: Asn=N

5=Glycine=Gly=G

Define ribonucleases with the example of titration curve:

Ribonucleases are enzymes that catalyze the degradation of RNA molecules. The titration curve of ribonucleases demonstrates changes in pH during titration, revealing optimal enzymatic activity. Understanding these curves aids in optimizing reaction conditions for RNA hydrolysis and related biochemical processes.

How alpha helix stabilize?

Alpha helices stabilize through hydrogen bonding between the carbonyl oxygen of one amino acid and the amide hydrogen of an amino acid four residues away. This regular pattern forms a stable helical structure, crucial for protein stability and function.

Define Perks of Quaternary structure:

The perks of quaternary protein structure lie in increased stability and functionality. Larger size enhances interactions among amino acid residues, making unfolding and refolding challenging. Multisubunit structures offer advantages like cooperativity in ligand binding, high-affinity binding sites for large molecules, and diverse subunit activities collaborating for specific functions, exemplified by enzyme complexes like Pyruvate Dehydrogenase.

Explain the D galactose:

D-galactose is a stereoisomer of glucose, belonging to the aldohexose family of sugars. It has a six-carbon structure and plays a role in various biological processes. Found naturally in lactose (milk sugar) and some plant polysaccharides, D-galactose undergoes metabolism in cells, contributing to energy production. Its configuration as a D isomer distinguishes it from its mirror-image L isomer.

What are the characteristics of positive charged amino acid R group?

Positive-charged amino acid R groups contain basic side chains, typically with amine groups. Notable examples include arginine, lysine, and histidine. These residues can accept protons, making them basic, and contribute to the overall positive charge of the protein. Their positively charged nature enables interactions with negatively charged molecules, such as phosphate groups in nucleic acids, facilitating various biochemical processes.

Write a note on Furfural compounds:

Furfural compounds are organic compounds containing a furan ring with an aldehyde functional group. Typically derived from plant biomass, furfurals are important in the production of biofuels and chemicals. Furfural itself is used in industrial applications, such as solvents and resins. These compounds contribute to the sustainable utilization of renewable resources in the chemical and energy sectors.